No history advantage in repeated environment-assisted qubit recovery

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Abstract

We compare current-outcome polar recovery with recovery conditioned on the complete measurement history for repeated uses of a qubit channel. The same four-outcome projective measurement is made on each fresh minimal environment. For every channel with full Choi rank and every positive use count, we prove that the separately optimized entanglement fidelities coincide and equal $(1 + \|Q\|_*^n)/2$, where Q polarizes the Kraus determinant. The key step is a quaternionic two-plane lemma that constructs, inside the complete real Takagi-optimal family, a Kraus frame with pairwise commuting effects. Thus retaining the outcome history gives no optimized fidelity advantage, including at three uses where fixed noncommuting measurements can first have complex branch traces.

Keywords: quantum feedback; environment-assisted error correction; qubit channel; entanglement fidelity; Kraus decomposition; Takagi factorization

1 Introduction

Measuring an environment after it interacts with a quantum system reveals a classical error record. Conditional operations on the system can then remove the unitary part of the observed error. This environment-assisted viewpoint connects quantum feedback and error correction without introducing a code subspace. For a fixed fine-grained channel decomposition, polar recovery is optimal at one step, and the optimal qubit recovery over several discrete steps can be represented by one final polar correction [1]. The latter representation makes the entire outcome history available to the recovery.

Rexiti, Khabbazi Oskouei, and Mancini recently compared such history-dependent ("Bayesian") recovery with a memoryless ('Markovian') strategy that applies the polar correction associated only with the current outcome [2]. In the version used here, equality is established for Kraus-rank-two channels and for a special rank-three family, while selected rank-three and rank-four cases are supported numerically. Their Conjecture 9 asserts equality for every qubit channel and every number of uses. The remaining issue is not the optimal recovery for a fixed decomposition. It is whether an environment measurement can be chosen that is simultaneously optimal for the history-dependent score and for memoryless polar recovery.

Three uses expose this issue sharply. A trace of two positive 2×2 matrices is real, whereas a trace of three positive matrices can be complex. Consequently, identities adequate at two uses do not by themselves close the three-use comparison. A generic environment measurement can indeed give a strict pointwise history advantage. A counterexample to the optimized conjecture, however, would require a gap between two separately optimized scores, not merely a gap at one nonoptimal measurement.

We prove that no such optimized gap exists for a qubit channel with positive-definite Choi matrix. More strongly, within the same fixed-measurement and fresh-environment model,

the optimized scores coincide for every positive integer number of uses. The proof has two ingredients. First, Takagi factorization identifies the global history-dependent optimum and leaves an entire real orthogonal family of optimal Kraus frames. Second, a new four-dimensional quadratic-form lemma selects a member of this family whose four effects commute. The lemma uses quaternionic left–right multiplication to find a two-plane on which three traceless quadratic forms are simultaneously trace neutral. Bases in that plane and its complement are then aligned by an intersection argument.

The result is deliberately restricted. The environment is fresh at each use, the same orthogonal projective measurement is repeated, the outcome record is perfect, and corrections are unitary polar recoveries. Adaptive environment measurements, additional-outcome positive-operator-valued measures, correlated uses, imperfect records, higher system dimension, and nonunitary recovery models are not covered.

2 Channel, measurement, and feedback scores

Let $\mathcal{E} : M_2(\mathbb{C}) \to M_2(\mathbb{C})$ be completely positive and trace preserving (CPTP), with positive-definite Choi matrix. Its minimal Kraus rank is four. Fix a minimal frame $A = (A_0, A_1, A_2, A_3)$,

$$\mathcal{E}(\rho) = \sum_{a=0}^{3} A_a \rho A_a^\dagger, \qquad \sum_{a=0}^{3} A_a^\dagger A_a = \mathbb{1}_2. \tag{1}$$

An orthogonal projective measurement of a minimal four-dimensional environment is parametrized by $U \in U(4)$ and produces

$$K_x(U) = \sum_{a=0}^{3} U_{xa} A_a, \qquad x = 0, 1, 2, 3. \tag{2}$$

The same U is used at every fresh, independent application of $\mathcal{E}$. Optimization over $U(4)$ removes dependence on the initially selected minimal frame.

Write the polar decomposition as $K_x = V_x P_x$, where $P_x = |K_x| = (K_x^\dagger K_x)^{1/2} \geq 0$. A partial isometry V_x can be extended unitarily on an unused kernel. Applying $V_x^\dagger$ after the current outcome gives the one-step corrected channel

$$\mathcal{M}_U(\rho) = \sum_x P_x \rho P_x, \qquad \sum_x P_x^2 = \mathbb{1}_2. \tag{3}$$

After n repetitions, its normalized entanglement fidelity is

$$F_n^{\mathrm{M}}(U) = \frac{1}{4} \sum_{x_1,\ldots,x_n} \left| \mathrm{tr}(P_{x_n} \cdots P_{x_1}) \right|^2 . \tag{4}$$

The modulus in Eq. (4) is essential for $n \geq 3$.

In the history-dependent protocol the correction at a given step may depend on every outcome observed so far. The optimal qubit score for a fixed fine-grained decomposition is equivalently obtained by delaying the polar recovery until the end [1]. In the present notation it is

$$F_n^{\mathrm{B}}(U) = \frac{1}{4} \sum_{x_1,\ldots,x_n} \left(\mathrm{tr} \left| K_{x_n} \cdots K_{x_1} \right| \right)^2 . \tag{5}$$

We compare the separately optimized quantities

$$F_n^{\mathrm{M},*}(\mathcal{E}) = \max_{U \in U(4)} F_n^{\mathrm{M}}(U), \qquad F_n^{\mathrm{B},*}(\mathcal{E}) = \max_{U \in U(4)} F_n^{\mathrm{B}}(U). \tag{6}$$

Because current-outcome polar recovery is an admissible history-dependent policy, $F_n^{\mathrm{M}}(U) \leq F_n^{\mathrm{B}}(U)$ for each fixed U.

3 Determinant reduction and the three-use obstruction

For every complex 2×2 matrix X,

$$(\operatorname{tr}|X|)^2 = \operatorname{tr}(X^\dagger X) + 2|\det X|. \tag{7}$$

Summing the first term over all length-n Kraus histories gives $\operatorname{tr}\mathbb{1}_2 = 2$, while determinants factor along a history. Hence, with

$$s(U) = \sum_x |\det K_x(U)|, \tag{8}$$

the fixed-measurement Bayesian score reduces exactly to

$$F_n^{\mathrm{B}}(U) = \frac{1 + s(U)^n}{2}. \tag{9}$$

The determinant of a linear combination of 2×2 matrices is a homogeneous quadratic form. There is therefore a complex symmetric matrix $Q = Q^{\mathsf{T}} \in M_4(\mathbb{C})$ such that

$$\det\left(\sum_a z_a A_a\right) = z^{\mathsf{T}} Q z. \tag{10}$$

Equation (2) gives

$$s(U) = \sum_x |(UQU^{\mathsf{T}})_{xx}| \le \|UQU^{\mathsf{T}}\|_* = \|Q\|_*, \tag{11}$$

where $\|Q\|_* = \operatorname{tr}\sqrt{Q^\dagger Q}$ is the trace norm. If $Q = V\Sigma V^{\mathsf{T}}$ is a Takagi factorization with $\Sigma = \operatorname{diag}(\sigma_0, \ldots, \sigma_3) \ge 0$, then $U = V^\dagger$ attains the bound. Thus

$$F_n^{\mathrm{B},*}(\mathcal{E}) = \frac{1 + \|Q\|_*^n}{2}. \tag{12}$$

Takagi decompositions play an analogous determinant-ensemble role in two-qubit concurrence problems [3]; Eq. (12) is not the missing part of the feedback comparison.

To see what is missing at three uses, expand each positive effect square root as

$$P_x = a_x \mathbb{1}_2 + \boldsymbol{b}_x \cdot \boldsymbol{\sigma}, \qquad a_x \in \mathbb{R}, \quad \boldsymbol{b}_x \in \mathbb{R}^3, \tag{13}$$

and set $G = \sum_x \boldsymbol{b}_x \boldsymbol{b}_x^{\mathsf{T}} \ge 0$. Trace preservation in Eq. (3) implies

$$s(U) = 1 - 2\operatorname{tr} G, \qquad R(U) = s(U) I_3 + 2G, \tag{14}$$

where $R(U)$ is the Bloch matrix of $\mathcal{M}_U$. Consequently

$$F_n^{\mathrm{M}}(U) = \frac{1 + \operatorname{tr} R(U)^n}{4}. \tag{15}$$

Let $\beta_1, \beta_2, \beta_3 \ge 0$ be the eigenvalues of G, $t = \sum_i \beta_i = (1 - s)/2$, $e_2 = \sum_{i<j} \beta_i \beta_j$, and $e_3 = \beta_1 \beta_2 \beta_3$. Direct expansion yields

$$F_3^{\mathrm{B}}(U) - F_3^{\mathrm{M}}(U) = 6\big[(s + t)e_2 - e_3\big] \ge 0. \tag{16}$$

Apart from the noiseless boundary, equality holds precisely when $\operatorname{rank} G \le 1$, or equivalently when all P_x commute. Thus a three-use proof requires a Kraus frame that both attains Eq. (11) and has commuting effects. Takagi factorization alone does not provide the second property.

4 A collinear-diagonal basis for three quadratic forms

We now prove the geometric statement that supplies the missing compatibility. It is distinct from simultaneous hollowization of a prescribed pair of matrices [4]: the output line is free to be chosen, and three traceless forms are treated in the special dimension four.

Lemma 4.1 (Collinear diagonals)**.** *Let H_1, H_2, H_3 be real symmetric 4×4 matrices with zero trace. There is an orthonormal basis $w_1, \dots, w_4$ of $\mathbb{R}^4$ such that the four vectors*

$$q(w_j) = (w_j^{\mathsf{T}} H_1 w_j, w_j^{\mathsf{T}} H_2 w_j, w_j^{\mathsf{T}} H_3 w_j) \in \mathbb{R}^3 \tag{17}$$

are collinear. Zero vectors are allowed.

Proof. Identify $\mathbb{R}^4$ with the quaternions $\mathbb{H}$ in the orthonormal basis $(1, i, j, k)$. For pure unit quaternions u, v, denote left and right multiplication by L_u and R_v. These real matrices are skew symmetric, square to $-I_4$, and commute with one another. Hence

$$\Gamma(u, v) = L_u R_v \tag{18}$$

is a symmetric orthogonal involution. It has zero trace, so

$$P(u, v) = \frac{I_4 + \Gamma(u, v)}{2} \tag{19}$$

is a rank-two orthogonal projection.

Put $(e_1, e_2, e_3) = (i, j, k)$. For a traceless symmetric H, define the real 3×3 matrix

$$C(H)_{ab} = \operatorname{tr}(H L_{e_a} R_{e_b}). \tag{20}$$

Writing pure quaternions in their three real coordinates and using $\operatorname{tr} H = 0$ gives

$$\operatorname{tr}\big(P(u, v) H\big) = \frac{1}{2} u^{\mathsf{T}} C(H) v. \tag{21}$$

For $C_k = C(H_k)$, consider the continuous cubic

$$g(u) = \det\left[C_1^{\mathsf{T}} u, C_2^{\mathsf{T}} u, C_3^{\mathsf{T}} u\right], \qquad u \in S^2. \tag{22}$$

It is odd. A path from u to $-u$ therefore contains a zero u_*. At that zero the three vectors $C_k^{\mathsf{T}} u_*$ span a subspace of dimension at most two. Choose a unit v_* perpendicular to their span. By Eq. (21),

$$\operatorname{tr}(P H_k) = 0, \qquad k = 1, 2, 3, \tag{23}$$

for $P = P(u_*, v_*)$. Since $\operatorname{tr} H_k = 0$, the same identities hold on $P^{\perp}$. We have therefore split $\mathbb{R}^4$ into two orthogonal two-planes on which all three compressed forms are trace neutral.

Choose an orthonormal basis (p_1, p_2) in the range of P. Each compressed form has the shape

$$\begin{pmatrix} a_k & b_k \\ b_k & -a_k \end{pmatrix}. \tag{24}$$

For $p(\theta) = \cos\theta\, p_1 + \sin\theta\, p_2$,

$$q(p(\theta)) = \boldsymbol{a} \cos(2\theta) + \boldsymbol{b} \sin(2\theta), \tag{25}$$

where $\boldsymbol{a} = (a_1, a_2, a_3)$ and $\boldsymbol{b} = (b_1, b_2, b_3)$. Its orthogonal partner has the negative q vector. The same description holds on $P^{\perp}$. Each pair therefore traces a centered ellipse contained in a linear subspace of $\mathbb{R}^3$ of dimension at most two.

If either ellipse map has rank at most one, its 3×2 defining map has a nonzero kernel. Choosing the corresponding angle makes both q vectors in that two-plane zero; any orthonormal pair in the other plane gives an opposite pair, so all four vectors are collinear. If both maps have rank two, their two-dimensional ranges in $\mathbb{R}^3$ intersect in a line. Choose angles whose images lie on that common line. The orthogonal partners again give the negatives. These cases include every possible degeneracy and prove the claim. □

5 A Takagi-optimal frame with commuting effects

The real freedom left after Takagi factorization is exactly large enough to apply Lemma 4.1. Start from $Q = V\Sigma V^{\mathsf{T}}$ and mix the original frame by $V^{\dagger}$, writing the resulting Kraus operators as $B_0, \ldots, B_3$. For a real vector $r \in \mathbb{R}^4$, put

$$K(r) = \sum_{j=0}^{3} r_j B_j. \tag{26}$$

Then

$$\det K(r) = r^{\mathsf{T}} \Sigma r \geq 0. \tag{27}$$

Consequently, for every $O \in O(4)$, the four rows r_x define a frame mixed by $U = OV^{\dagger}$ for which

$$\sum_x |\det K(r_x)| = \sum_{x,j} \sigma_j O_{xj}^2 = \operatorname{tr} \Sigma = \|Q\|_*. \tag{28}$$

Every member of this real orthogonal family is globally optimal for the Bayesian score.

Expand the effect of $K(r)$ in the Pauli basis:

$$K(r)^{\dagger} K(r) = \alpha(r) \mathbb{1}_2 + \sum_{k=1}^{3} (r^{\mathsf{T}} H_k r) \sigma_k. \tag{29}$$

Because r is real, each coefficient is a real quadratic form and the H_k may be chosen real symmetric. More explicitly,

$$r^{\mathsf{T}} H_k r = \frac{1}{2} \operatorname{tr}[\sigma_k K(r)^{\dagger} K(r)]. \tag{30}$$

Completeness of the Takagi-mixed frame gives

$$\operatorname{tr}_{\mathbb{R}^4} H_k = \frac{1}{2} \operatorname{tr} \left[\sigma_k \sum_j B_j^{\dagger} B_j \right] = \frac{1}{2} \operatorname{tr} \sigma_k = 0. \tag{31}$$

Apply Lemma 4.1 to (H_1, H_2, H_3) and use its basis vectors as the rows of O. Equation (29) then says that the Bloch vectors of the four effects $K(r_x)^{\dagger} K(r_x)$ lie on one line. Two qubit Hermitian matrices commute precisely when their Bloch vectors are collinear. The four effects, and hence their positive square roots, commute. Combined with Eq. (28), this proves the following.

Proposition 5.1 (Optimal commuting frame)**.** *Every full-Choi-rank qubit CPTP map has a four-outcome orthogonal environment measurement that simultaneously satisfies*

1. $\sum_x |\det K_x| = \|Q\|_*$, *and*
2. $[K_x^{\dagger} K_x, K_y^{\dagger} K_y] = 0$ *for all* x, y.

No nondegeneracy of the Takagi singular values or of the individual outcome operators was used. In particular, zero or singular outcomes do not obstruct the construction.

6 Equality for every use count

Theorem 6.1 (No optimized history advantage)**.** *Let $\mathcal{E}$ be a qubit CPTP map with positive-definite Choi matrix. Use the same four-outcome orthogonal projective measurement on every fresh minimal environment. For every positive integer n, the separately optimized entanglement fidelities of current-outcome polar recovery and full-history optimal recovery satisfy*

$$\boxed{F_n^{\mathrm{M},*}(\mathcal{E}) = F_n^{\mathrm{B},*}(\mathcal{E}) = \frac{1 + \|Q\|_*^n}{2}.} \tag{32}$$

In particular, the optimized three-use gap is zero.

Proof. Choose the frame from Proposition 5.1 and diagonalize its commuting square roots in one system basis:

$$P_x = |K_x| = \begin{pmatrix} p_x & 0 \\ 0 & q_x \end{pmatrix}, \qquad p_x, q_x \geq 0. \tag{33}$$

Trace preservation and determinant optimality give

$$\sum_x p_x^2 = \sum_x q_x^2 = 1, \qquad \sum_x p_x q_x = \sum_x |\det K_x| = \|Q\|_*. \tag{34}$$

The corrected channel $\mathcal{M}_U$ preserves the longitudinal Bloch component and multiplies both transverse components by $\|Q\|_*$. Its Bloch eigenvalues are therefore $(1, \|Q\|_*, \|Q\|_*)$, and Eq. (15) gives

$$F_n^{\mathrm{M}}(U) = \frac{1 + \|Q\|_*^n}{2}. \tag{35}$$

By Eqs. (9) and (28), the Bayesian score at the same U has exactly this value, which is its global optimum in Eq. (12). Finally, $F_n^{\mathrm{M},*} \leq F_n^{\mathrm{B},*}$ because the memoryless policy is a special history-dependent policy, whereas the constructed frame shows the reverse inequality at the common optimum. This proves Eq. (32). □

The theorem does not say that history is useless at a prescribed environment measurement. Equation (16) shows that a noncommuting fixed frame can have a strict three-use advantage. Rather, it says that an optimal measurement can always be chosen from the Takagi family so that the memoryless corrected channel already reaches the global history-dependent ceiling. This separation between pointwise and optimized statements is essential to the physical interpretation.

7 Deterministic numerical checks

The argument above is finite dimensional and exact; numerical evidence is not used to infer Theorem 6.1. Deterministic scripts accompanying this article implement the quaternionic construction and independently enumerate Kraus histories as diagnostic checks. Twelve arbitrary triples of traceless real symmetric forms and eight nonunital channels with strictly positive Choi spectra were tested. The latter used $n = 1, \ldots, 7$ transfer calculations and an explicit $4^3 = 64$-history calculation at $n = 3$. Table 1 reports the largest residual in each class.

The smallest Choi eigenvalue among these eight channel instances was 3.662×10^{-3}. The same package reproduces the published special rank-three baseline of Ref. [2] for three interior parameters and $n = 1, 2, 3, 5$, with largest residual 1.887×10^{-15}. It also checks the one-step polar-recovery result of Ref. [1] and verifies the complex-modulus convention in Eq. (4) by a direct normalized Choi–Bell contraction.

Table 1: Floating-point diagnostics for the exact construction. These residuals check the implementation and are not substitutes for the proof.

Diagnostic	largest residual
generic three-form construction	1.639×10^{-15}
Takagi diagonalization	1.364×10^{-15}
determinant optimum $s = \|Q\|_*$	4.441×10^{-16}
effect commutators	1.316×10^{-16}
score identity, $n = 1, \ldots, 7$	1.554×10^{-15}
explicit 64-history identity, $n = 3$	3.331×10^{-15}

8 Discussion and limitations

The trace norm of the determinant polarization Q controls the common optimum, but the main structural result is not the Takagi bound itself. The nontrivial step is compatibility: among all determinant-optimal measurements, there is one whose four effects commute. Lemma 4.1 establishes this without assuming a unital channel, distinct Takagi singular values, or invertible outcome operators. It also explains why a generic simultaneous hollowization result for a fixed pair of matrices is insufficient: the proof must exploit both four real Kraus coordinates and the freedom to select the surviving common Bloch axis.

At three uses, noncommuting positive factors may have a complex trace, and a fixed measurement can make full outcome history operationally useful. The optimization over the projective environment basis nevertheless removes this advantage. The all-n extension follows because the constructed channel has the exact Bloch spectrum $(1, s, s)$; it is not a perturbative extrapolation from the three-use case.

Several nearby questions remain open outside the theorem's domain. An adaptive choice of environment basis changes the admissible optimization. Extra-outcome nonprojective measurements are not parametrized by $U(4)$. Correlated environments invalidate the product-history factorization behind Eq. (9). Nonunitary correcting channels, imperfect records, and higher-dimensional systems require different optimality arguments. No experimental performance or implementation-cost claim is made. Finally, the literature comparison supporting priority is bounded rather than exhaustive and requires human review before submission.

Statements and declarations

Data and code availability. There are no experimental data. The reproducibility package accompanying this draft contains the Python source, fixed-seed logs, environment requirements, and commands for the published-baseline and construction checks described in Sec. 7.

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